

SURIYAN DHANAPAL

AI / GenAI Engineer | LLM, RAG & Voice AI Infrastructure

Coimbatore, India | suriyandhanapal@gmail.com | +91 81100 44608

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI engineer with 2 years building production GenAI systems in Python/FastAPI — LLM provider infrastructure, RAG retrieval, agentic workflows and real-time voice pipelines. Shipped a provider layer unifying **25+** LLM, STT and TTS services behind one connector interface, normalizing incompatible function-calling schemas across vendors; built multi-tenant RAG over pgvector and Weaviate; and designed human-in-the-loop approval gates governing how much an agent may do unsupervised. Equally comfortable deciding where an LLM does *not* belong.

TECHNICAL SKILLS

AI & LLM: LLM integration (OpenAI, Google Vertex AI/Gemini, AWS Bedrock, Groq, Sarvam), AI agents & agentic workflows, LangGraph, tool calling / function-calling schema design, prompt design, RAG & retrieval, chunking & embeddings, semantic search, human-in-the-loop approval gates, provider regression testing

Vector & Data: pgvector, Weaviate, PostgreSQL, MongoDB, Redis, Cassandra, SQLAlchemy, schema design, SQL

Speech & Multimodal: STT, TTS, STS pipelines (Whisper, ElevenLabs, Sarvam), real-time voice agents over WebRTC/SIP, vision-model document extraction, OCR

Backend: Python, FastAPI, Node.js, Express, REST API design, microservices, gRPC, WebSockets, async workers (Redis/RQ), OAuth 2.0, JWT, RBAC, multi-tenancy, idempotency & retry design, webhook delivery

Infra: Docker, Linux/shell, nginx, GCP, AWS (S3, Bedrock), GitHub Actions (CI/CD)

Frontend: React, Next.js, Redux Toolkit, Vite, Tailwind CSS, React Native (Expo)

Tools: Git, Claude Code

PROFESSIONAL EXPERIENCE

AI / Software Engineer, Mindgryd Software Private Limited

Jun 2025 – Aug 2026

AI agent platform · Coimbatore, India

Tech: Python, FastAPI, LangGraph, embeddings, RAG, Weaviate, pgvector, PostgreSQL, OpenAI, Vertex AI/Gemini, AWS Bedrock, Groq, Whisper, ElevenLabs, Sarvam, WebRTC, SIP, gRPC, Redis, MongoDB, Cassandra, Docker, GCP · Next.js, TypeScript, Vite + React

- Integrated **25+** LLM, STT, TTS and STS providers (OpenAI, Gemini, Vertex AI, Groq, Sarvam, Whisper, ElevenLabs) behind a single unified connector interface on FlowCat, the team's open-sourced real-time media engine, cutting new-provider integration from **days to hours** and serving **~100** live voice-agent calls/day in pilot.
- Normalized provider divergence behind that one interface — incompatible function-calling and tool-call schemas, gRPC streaming transports, API-key resolution — and built provider-aware regression flows that caught breaking vendor changes before release rather than on live calls.
- Built the RAG retrieval pipeline behind the AI memory and context platform: chunked and embedded ingested documents into PostgreSQL/pgvector, with metadata-scoped retrieval so every query stayed inside the requesting organization's own corpus — tenant isolation enforced at the retrieval layer, not just the API.
- Built a workspace-aware agent recommendation system over semantic search: embedded onboarding context, indexed it in Weaviate, and retrieved and ranked from a pool of **several hundred** personalized agents inline on every search.
- Designed the agent autonomy model — human-in-the-loop approval gates governing how far an agent could act before an action required sign-off — and the execution and schema design behind the workflow automation platform.
- Built the knowledge-ingestion layer feeding retrieval: **8** connectors including Google Drive, Dropbox, Notion and SharePoint, with per-user OAuth token storage and refresh and content normalization across heterogeneous source formats.
- Debugged provider-specific failures on live voice calls — streaming disconnects, tool-call mapping mismatches, key resolution — and built a real-time Vite + React dashboard surfacing call status, provider health and test-flow results, which became the team's primary triage tool.
- Owned the surrounding platform engineering: OAuth 2.0 authentication with Google/GitHub/Microsoft, JWT sessions, RBAC and automatic organization provisioning; an HMAC-signed, at-least-once webhook delivery service handling **hundreds** of events/day; and the Razorpay/PayPal payments layer.

Software Engineer, Hifresh Agro Private Limited

Sep 2024 – Jun 2025

Agro-tech product company · Coimbatore, India (same founding team; moved to Mindgryd when it was spun up)

Tech: Node.js/Express, REST APIs, MongoDB, React Native (Expo), Redux, Zoho Books API, response caching

- Built and owned a full-stack internal delivery-management application used daily by the company's delivery team and handling ~**150** orders/day — React Native/Expo client on an extended Node.js/Express backend, with APIs for delivery assignment, billing and Zoho Books invoicing.
- Added response caching and structured logging to the backend's highest-traffic endpoints, cutting response time from ~**2s** to **under ~300 ms**.

CONTRACT / PROJECT WORK

Thavasi Scrutiny — Indian-Customs Compliance Platform, Cooper Compass

Aug 2026 – Present

IIT Bombay-incubated startup · thavasi.coopercompass.com

Tech: Python, FastAPI, OpenAI (extraction assist only), vision-model OCR, pdfplumber, camelot, pytesseract, SQLAlchemy, PostgreSQL, Redis/RQ, JWT, Docker, nginx · React, TypeScript, Vite

- **Kept the LLM out of the decision path by design:** it assists extraction on layouts the deterministic parsers miss and explains flagged fields, while matching and verdicts stay pure functions of the inputs and a versioned rule config — so an audit re-run months later reproduces the same result. Non-negotiable in a regulatory domain where a hallucinated verdict is a customs penalty.
- Built a hybrid extraction pipeline, cheapest-path-first: pdfplumber, camelot, pytesseract and openpyxl do the reading, with a **vision model gated to genuinely scanned pages** so inference cost is spent only where deterministic parsing actually fails. Runs in Redis/RQ workers so uploads return immediately and the client polls for completion.
- Designed the deterministic scrutiny engine comparing **18** catalogued customs fields, **10** of them zero-tolerance, across air, sea, road and rail — resolving transport mode before comparison so mode-specific fields are never demanded of the wrong shipment type.
- Shipped the platform end to end for Customs House Agents: a filer uploads a draft checklist plus source documents and receives a field-by-field discrepancy report, every value traced back to the document, page and cell it was read from. Piloted with real Customs House Agents against live filings.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering

2019 – 2023

Dr. N.G.P. Institute of Technology